John Gomes Combustion HW 2

4) $\frac{dO_3}{dt} = k_{exp}[O_3^2/O_2]$

$M + O_3 \xrightarrow{k_1} O_2 + O + M$

$M + O + O_2 \xrightarrow{k_2} O_3 + M$

$O + O_3 \xrightarrow{k_3} 2O_2$

a) $dO_3/dt = -k_1(O_3)(M) + k_2(O_2)(O)(M) - k_3(O_3)(O)$

$dO/dt = k_1(O_3)(M) - k_2(O_2)(O)(M) - k_3(O_3)(O)$

$\frac{dR}{dt} = 0$ (radical), rewrite for concentration (O), $k_2 = k_3 = k_{eq}$

$$(O) = \frac{k_1(O_3)(M)}{k_{eq}((O_2)(M) + (O_3))}$$

plug into dO_3/dt:

$$\frac{dO_3}{dt} = -k_1(O_3)(M) + \cancel{k_{eq}}(O_2)(M) \cdot \frac{k_1(O_3)(M)}{\cancel{k_{eq}}((O_2)(M)+(O_3))} - \cancel{k_{eq}}(O_3)\frac{k_1(O_3)(M)}{\cancel{k_{eq}}((O_2)(M)+O_3)}$$

common denominator:

$$dO_3/dt = k_1(O_3)(M)\frac{[-\cancel{(O_2)(M)} - O_3 + \cancel{(O_2)(M)} - (O_3)]}{(O_2)(M)+O_3}$$

$$dO_3/dt = \frac{k_1(O_3)(M)[2(O_3)]}{(O_2)(M)+O_3} = \frac{k_1(O_3)(M)(2\cdot(O_3))}{(O_2)(M)} + \frac{\text{"}\cdot\text{"}}{O_3}$$

$$\boxed{dO_3/dt = \frac{-2k_1(O_3)^2}{O_2} - 2k_1(O_3)(M)}$$

based on this, $\boxed{O_2 >> O_3}$ as this would deem the second term negligible, leaving us with only the first term, which is very similar to the experimental result.

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4) b) Based on $\frac{dO_3}{dt}$, $k_{exp}\frac{(O_3^2)}{(O_2)}$, $n = 2 + -1 = 1$

$\boxed{\text{overall order} = 1}$

c) assume $O_2 >> O_3$, but $k_2 \neq k_3$

$$\frac{dO_3}{dt} = \frac{-2(M)(O_3^2)\ k_1 k_3}{\cancel{(O_3)k_3} + (M)(O_2)k_2} = \frac{-2(M)(O_3^2)(k_1)(k_3)}{(M)(O_2)k_2}$$

k_1 and k_2 are the same reaction in opposite direction, therefore:

$K_c = k_f / k_b = k_1 / k_2$, which can be found as a function of T in tables...

$$\frac{dO_3}{dt} = \frac{-2\cancel{(M)}\cancel{(O_3^2)}\ k_c \cdot k_3}{\cancel{O_2}\cancel{(M)}} = k_{exp}\frac{\cancel{(O_3^2)}}{\cancel{O_2}}$$

also a function of T

$$(-2)k_c k_3 = k_{exp}$$

$$\boxed{k_3 = \frac{k_{exp}}{-2K_c}}$$

you can find k_3

d) you could vary the concentration of O, O_2, O_3 to manipulate the k values, to say make k_c dominate, thus making it the overall reaction constant

e: could you continue to compare experimental rate of decay of ozone to analytical at various conditions?

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9)

$H_2 + OH \xrightarrow{k_f} H_2O + H$ $\qquad$ $T = 1000\ K$

$k_f = AT^n \exp(-E/RT)$ $\qquad$ using tables:

$A = 2.16 \times 10^8$

$n = 1.5$

$E = 14.35$

$R = 0.008123\ kJ/mol\,K$

$k_f = 2.16 \times 10^8 (1000)^{1.5} \exp\left(\frac{-14.35}{0.008123 \cdot 1000}\right)$

$$\boxed{k_f = 1.2 \times 10^{12} \left[\frac{cm^3}{s \cdot mol}\right]}$$

find units for k_f:

given $H_2 + OH \xrightarrow{k_f} H_2O + H$

$\frac{d[OH]}{dt} = -k_f [OH][H_2]$

$k[A][A]$

second order:

$k = [cm^3/s \cdot mol]$

$K_c = k_f / k_b$

$k_b = k_f / K_c$

$K_c = \frac{K_p(H_2O) \cdot K_p(H)}{K_p(OH)\, K_p(H_2)}$

from tables:

	$\log K_f$
H_2O	+0.016
H	−8.6
OH	−1.22
H	0

$K_c = 432.5$

$k_b = 1.2 \times 10^{12} / 432.5 \Rightarrow \boxed{k_b = 2.8 \times 10^9 \left[\frac{cm^3}{s \cdot mol}\right]}$

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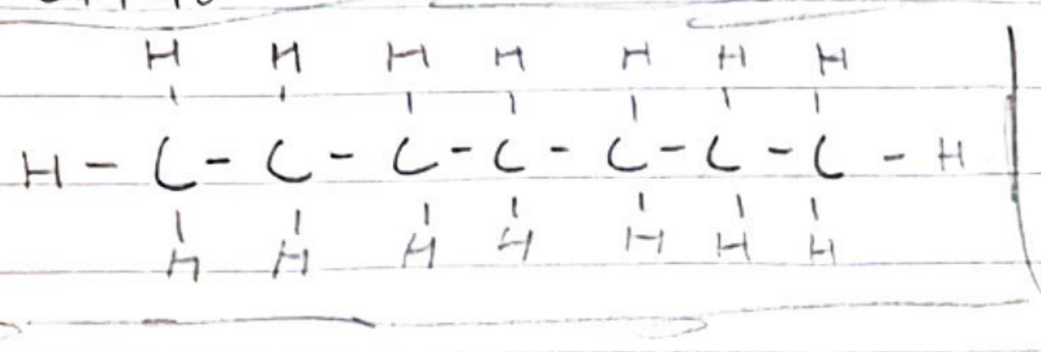
3)

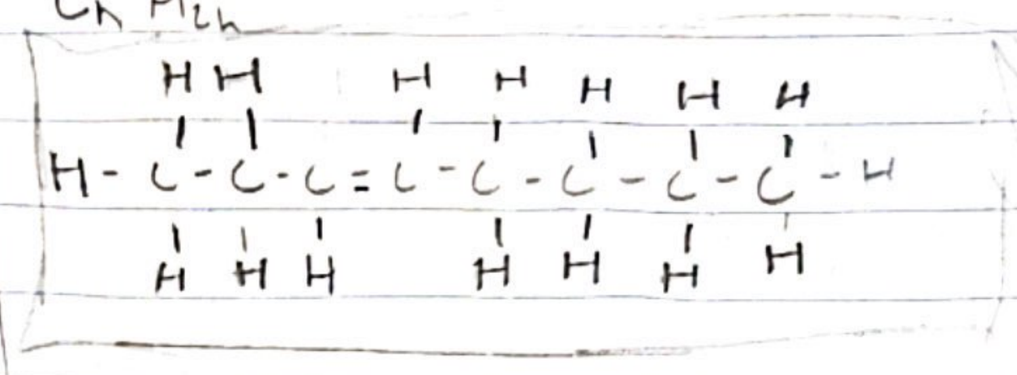
heptane

C_7H_{16}



3-octene $n=8 \Rightarrow C_8H_{16}$

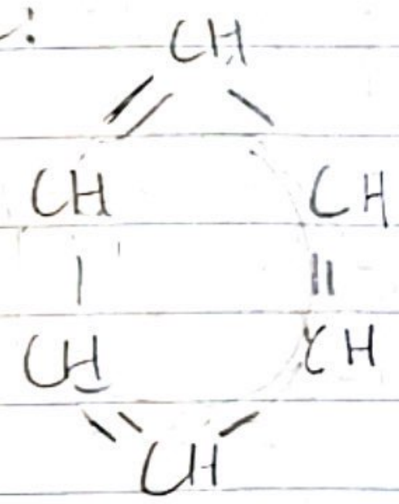
C_nH_{2n}



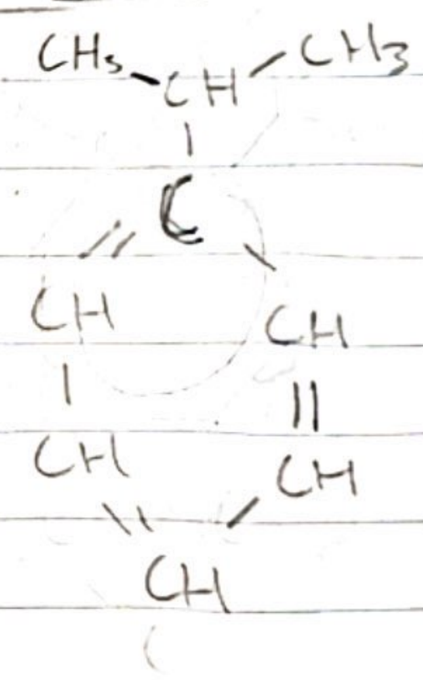
isopropyl benzene $(C_6H_5 - CH(CH_3)_2$

C

benzene:



isopropyl benzene:



C

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7)

$k_2(\alpha-1)M \gg k_4(M) + k_5(O_2)(M)$

eqn (3.8):

steady state concentration:

$$R = \frac{k_1(M)}{k_4(M) + k_5(O_2)(M) + k_6 - k_2(\alpha-1)(M)}$$

$M \xrightarrow{k_1} R$ initiation

$R + M \xrightarrow{k_2} \alpha R + M'$ propagation

$R + M \xrightarrow{k_3} P + R$ propagation

$R + M \xrightarrow{k_4} R + I$ termination

$R + O_2 + M \xrightarrow{k_5} RO_2 + M$ 3O termination

$R \xrightarrow{k_6} I'$ surface termination

if k_2 term >> k_4, k_5 terms, this shows steps 4/5 are not significant in the concentration of Radicals, R (H radicals)

Only the initiation, initial propagation & surface termination (forms intermediates) are significant to the radical concentration

Depending on the k values of k_0, k_1, k_2, this could lead to a lot of branching as there would be a high radical concentration - an explosive system

this makes sense as both (4) & (5) are terminating steps, which turn R into something else; RO_2 (less reactive), I, P

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8) TEL is a good anti-knocking agent because its ability to reduce the number of radicals ☆

knocking occurs when premature explosions happen in the piston of an engine

TEL forms non-crucial intermediates, lead oxide. This is step 6 in the previous problem, and as shown in 3.8 a k_6 term in the denom indicates less radicals

without (TEL) this step 6 would not happen as much. fortnall

☆ $\alpha_{crit} = 1 + \frac{100 + k_6}{100}$, thus α_{crit} increases with k_6 increasing

without TEL: $\alpha_{prod} > \alpha_{crit}$, which leads to a generative explosive system.

With TEL $\alpha_{prod} < \alpha_{crit}$, less premature explosions